

NEUMAYER STATION III, Antarctica

WMO: 890020

Lat: 70.676S

Lon: 8.274W

Elev: 50

StdP: 100.73

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-40.7	-38.8	-43.7	0.1	-40.3	-41.8	0.1	-38.6	29.1	-14.7	26.8	-17.5	4.1	150	1.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.3	-0.1	-1.0	-0.7	-1.5	-1.3	-2.0	-0.7	-0.3	-1.3	-0.8	-1.9	-1.3	7.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
-1.2	3.4	-0.6	-1.9	3.2	-1.1	-2.7	3.0	-1.6	8.5	-0.2	7.5	-0.7	6.4	-1.3	0.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.3	23.8	21.3	DB	-43.8	0.8	2.4	0.9	-45.5	1.5	-46.9	2.0	-48.3	2.5	-50.0	3.1	(4)
(5)			WB	-43.4	-0.2	1.6	0.5	-44.5	0.2	-45.4	0.5	-46.3	0.8	-47.5	1.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-16.4	-4.7	-8.5	-13.8	-18.3	-20.6	-23.5	-25.4	-24.5	-23.7	-17.7	-10.1	-5.1	(6)	
	DBStd	9.34	2.58	3.78	5.24	6.68	7.63	6.95	6.96	7.28	6.79	5.56	3.80	2.51	(7)	
	HDD10.0	9621	457	517	739	848	948	1004	1096	1070	1011	860	603	468	(8)	
	HDD18.3	12662	715	750	997	1098	1206	1254	1354	1328	1261	1118	853	727	(9)	
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	8.9	6.5	7.8	9.0	9.4	9.9	9.1	9.0	10.2	9.6	9.7	9.1	7.4	(14)	
(15) Precipitation	PrecAvg	409	33	32	35	32	40	26	25	32	29	38	49	39	(15)	
	PrecMax	586	72	70	88	62	74	72	60	52	75	79	105	86	(16)	
	PrecMin	271	7	8	10	11	11	8	4	11	7	4	14	10	(17)	
	PrecStd	69	15	16	19	15	14	15	12	13	15	18	23	19	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.5	0.0	-2.5	-3.3	-3.8	-7.1	-8.5	-7.0	-6.8	-4.3	-1.7	0.9	(19)	
		MCWB	-0.6	-0.6	-2.9	-3.6	-4.2	-7.5	-8.8	-7.5	-7.3	-4.9	-2.4	-0.4	(20)	
	2%	DB	0.0	-1.0	-4.1	-5.7	-6.7	-9.3	-11.2	-9.9	-10.8	-6.9	-2.9	-0.3	(21)	
		MCWB	-0.8	-1.5	-4.5	-6.2	-7.0	-9.6	-11.6	-10.2	-11.4	-7.4	-3.6	-1.4	(22)	
	5%	DB	-0.4	-2.1	-5.5	-7.8	-8.7	-11.8	-13.6	-12.2	-12.8	-8.7	-3.8	-0.9	(23)	
		MCWB	-1.2	-2.7	-6.1	-8.3	-9.2	-12.3	-14.0	-12.5	-13.1	-9.2	-4.4	-1.8	(24)	
	10%	DB	-0.9	-3.1	-6.8	-9.4	-10.3	-13.5	-15.6	-14.4	-14.7	-10.2	-4.7	-1.5	(25)	
		MCWB	-1.7	-3.7	-7.4	-9.9	-10.8	-13.9	-16.0	-14.8	-15.1	-10.7	-5.2	-2.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	-0.4	-2.7	-3.5	-4.1	-7.5	-8.8	-7.3	-7.1	-4.7	-2.1	-0.3	(27)	
		MCDB	0.2	-0.1	-2.6	-3.4	-3.9	-7.2	-8.6	-7.0	-6.7	-4.5	-1.9	0.4	(28)	
	2%	WB	-0.5	-1.5	-4.5	-6.3	-6.9	-9.6	-11.6	-10.2	-11.3	-7.4	-3.4	-1.1	(29)	
		MCDB	-0.2	-1.1	-4.2	-5.9	-6.8	-9.3	-11.2	-9.9	-11.0	-7.0	-3.1	-0.5	(30)	
	5%	WB	-1.0	-2.6	-6.0	-8.2	-9.1	-12.2	-13.9	-12.6	-13.1	-9.2	-4.3	-1.7	(31)	
		MCDB	-0.5	-2.1	-5.6	-7.8	-8.7	-11.9	-13.6	-12.2	-12.8	-8.8	-3.9	-1.1	(32)	
	10%	WB	-1.6	-3.6	-7.4	-9.9	-10.8	-13.9	-16.0	-14.7	-15.1	-10.7	-5.2	-2.2	(33)	
		MCDB	-1.0	-3.1	-6.9	-9.4	-10.4	-13.5	-15.7	-14.4	-14.7	-10.3	-4.7	-1.6	(34)	
(35) Mean Daily Temperature Range	MDBR		5.3	6.3	6.6	7.2	7.2	7.9	7.6	7.1	7.2	6.7	6.2	5.1	(35)	
	5% DB	MCDBR	4.5	5.4	5.5	6.9	6.5	7.6	7.3	5.7	6.7	6.2	5.7	4.5	(36)	
		MCWBR	4.2	5.2	5.5	6.8	6.4	7.5	7.2	5.6	6.6	6.0	5.4	4.0	(37)	
	5% WB	MCDBR	3.9	5.0	5.3	6.8	6.4	7.7	7.3	5.5	6.7	5.9	5.3	4.2	(38)	
		MCWBR	3.7	4.8	5.3	6.8	6.4	7.6	7.1	5.4	6.7	5.8	5.1	3.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.261	0.248	0.219	0.159	N/A	N/A	N/A	0.153	0.198	0.221	0.241	0.259	(40)	
	taud		2.257	2.297	2.282	2.053	N/A	N/A	N/A	1.998	2.197	2.211	2.226	2.227	(41)	
	Ebn at Noon		977	926	836	591	0	0	0	576	849	954	998	996	(42)	
	Edh at Noon		110	90	64	31	0	0	0	31	67	96	113	119	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.71	4.35	2.07	0.54	0.00	0.00	0.00	0.24	1.54	3.82	6.52	7.81	(44)	
	RadStd		0.42	0.35	0.20	0.05	0.00	0.00	0.00	0.02	0.10	0.18	0.26	0.43	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-1.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page